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Word Embeddings

Transform words into numerical vectors that capture meaning, revealing the semantic relationships between words in your model.

You will need

- your completed bigram model grid (including context columns if you have them)
- another empty grid (same size as your bigram model)
- pen, paper & dice as per *Grid Generation*

Your goal

To create a similarity matrix (another square grid) which captures how similar (or different) all the words in your bigram model are. **Stretch goal**: create a visual representation of this similarity matrix.

Key idea

Each word's row in your model is its embedding under that model — a numerical fingerprint that captures meaning through context. Distances between words reveal grammatical and semantic relationships. Similar words have similar embeddings.

Algorithm

For this algorithm you'll need two grids: your original *bigram model* grid and a new *embedding distance* grid (with the same words as row/column headers, but otherwise blank to start with).

1. for the first row and second row in the bigram model, add up the total of the absolute differences between corresponding cells in the two rows and write it in the empty cell for that word pair in the embedding distance grid
2. fill out the embedding distance grid by repeating step 1 for all different pairs of rows in the bigram model grid

Example

Original text: *"See Spot. Spot runs."*

Completed bigram model grid:

	see	spot	.	runs
see		|		
spot			|	|
.		|		
runs			|	

The embedding distance between the first two rows (see and spot) is the sum of the absolute differences between corresponding elements (0 for blank cells):

$$\begin{aligned} d(\text{see}, \text{spot}) &= |0-0| + |1-0| + |0-1| + |0-1| \\ &= 0 + 1 + 1 + 1 \\ &= 3 \end{aligned}$$

Put this distance in the embedding distance grid (note diagonals are already pre-filled with 0 as well):

	see	spot	.	runs
see	0	**3**		
spot		0		
.			0	
runs				0

Complete embedding distance grid (no need to fill out the bottom triangle — the embedding distance is symmetric):

	see	spot	.	runs
see	0	3	0	2
spot		0	3	2
.			0	2
runs				0

The distances show that see and . have identical embeddings (distance = 0), while see and spot are quite different (distance = 3).